Yuen-Hei Yeung

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Research interests: continual and self-improving learning for embodied agents; world models for perception and planning

Education

New York University (Courant Institute of Mathematical Sciences) — New York, United States
Master of Science in Computer Science, GPA 3.89/4.00 — Sep 2024 – Dec 2025

City University of Hong Kong — Hong Kong S.A.R., China
Bachelor of Science in Computer Science, GPA 3.52/4.30 — Sep 2018 – May 2022

Publications

B. Wang, **Y.H. Yeung***, V.R.S. Devarakonda*, X. Xiao, *Demonstration Synthesis from a Single Scan via Gaussian Splatting for Visuomotor Policy Learning* (under review at ICRA 2027) — arXiv Sep 2026

B. Wang, T. Xu, D. Kosukhin, **Y.H. Yeung**, Y. Lu, X. Xiao, *Learning Scene-Aware Humanoid Locomotion through 3D Clutter from Immersive Human Demonstrations* (under review at ICRA 2027) — arXiv Sep 2026

Y.H. Yeung, C. Hoang, Z. Zhao, M. Ren, *Canonical Observation Pretraining for Visuomotor Control* (under review at ICLR 2027) — CVPR 2026 Workshop Jun 2026

Y.H. Yeung, X.H. Zhu, Z.J. Chen, H. Chen, D. Su, S. Fan, *Method of Automatically Determining Sensor Placement in a Target Environment* (US 2025/0244452 A1; also WO, CN, HK) — US Patent Jul 2025

J. Lin*, **Y.H. Yeung***, R.W.H. Lau, *Leveraging RGB-D Data with Cross-Modal Context Mining for Glass Surface Detection* — AAAI Feb 2025

Y. Wang, H. Lv, Q.W. Teo, R. Lei, A.B. Gopal, W.O. Ouyang, **Y.H. Yeung**, T.J.C. Tan, D. Choi, I.R. Shen, X. Chen, C.S. Graham, N.C. Wu, *An Explainable Language Model for Antibody Specificity Prediction Using Curated Influenza Hemagglutinin Antibodies* — Immunity Oct 2024

J. Lin*, **Y.H. Yeung***, R.W.H. Lau, *Exploiting Semantic Relations for Glass Surface Detection* — NeurIPS Nov 2022

Research Experience

Agentic Learning AI Lab, NYU Courant Institute — New York, United States
Research Assistant under Prof. Mengye Ren — Jan 2025 – Present

- Designed *Canonical*, self-supervised pretraining for imitation-learning policies enabling **zero-shot, calibration-free deployment** under camera-viewpoint shifts; cut seen-to-unseen success drop by over 50% vs. baselines across simulated benchmarks and **real xArm robot** (CVPR 2026 Workshop; submitted to ICLR 2027)
- Implemented goal-conditioned policy warm starts for gradient-descent and CEM planners in lab's **JEPA world-model** framework (jepa-agent.org); with gradient descent, **5-step warm starts beat 100-step cold planning** on every task seed across PushT, PointMaze and Cube at **1/10–1/24 of wall-clock** (PushT: 61% → 88% with world-model data)

RobotiXX Lab, George Mason University — (Remote) Virginia, United States
Research Assistant under Prof. Xuesu Xiao — May 2026 – Sep 2026

- Trained **diffusion-transformer world model** predicting object tilt during grasp closure and slip during lift for *GaussianFactory*, which synthesizes visuomotor-policy demonstrations from single 3D Gaussian Splatting scan without teleoperation (arXiv 2026)
- Trained RL motion-tracking **teacher policies** and **DAgger** student policy for *MTC*, scene-aware humanoid locomotion through 3D clutter from VR human demonstrations, reaching **70–98%** simulated success across 5 traversal tasks (crawl: **81%** vs. 0% for CAT baseline) (arXiv 2026)

NVIDIA AI Technology Center — Hong Kong S.A.R., China
Research Assistant under Prof. Charles K.C. Cheung — Jun 2023 – Aug 2023

- Extended SVD-based **parameter-efficient transfer learning** from CNNs to vision transformers; improved ImageNet top-1 accuracy by **2%** while tuning **10x fewer parameters** than full fine-tuning

Wu Lab, School of Molecular and Cellular Biology, UIUC Illinois, United States
Research Assistant under Prof. Nicholas C. Wu Jul 2022 – Dec 2022

- Trained antibody-specificity classifiers on ∼5.5k curated influenza hemagglutinin antibodies for head-vs.-stem prediction, reaching **0.61 F1** with **kNN, Naive Bayes** and **XGBoost** ensemble (Immunity 2024)

Department of Computer Science, City University of Hong Kong Hong Kong S.A.R., China
Research Assistant under Prof. Rynson W.H. Lau Apr 2021 – Jun 2022

- Co-first author on two **glass-surface detection** papers (NeurIPS 2022, AAAI 2025): proposed transformer models leveraging semantic relations and cross-modal RGB-D context, reducing MAE by **42.6%** vs. prior SOTA; built two benchmarks (4.5k labeled images)

Industry Experience

Philo Homes (Startup; https://philo.homes) (Remote) Michigan, United States
CV Team Lead (Oct 2026 – Present); *Machine Learning Engineer* (Oct 2025 – Oct 2026) Oct 2025 – Present

- Built end-to-end **video-to-3D room reconstruction** pipeline for consumer interior-design app: **Depth Anything 3** joint depth/pose estimation, SAM 3 semantic segmentation, structural plane optimization, axis alignment, metric room-dimension estimation (**<10% error** for most residential rooms), and furnished/empty-room meshing
- Lead small CV team developing 3D room reconstruction and **LiDAR-based scanning** pipelines; designed method to **detect and seal unscanned regions**, producing complete room meshes supporting colleagues' downstream furniture editing and style-matched recommendation features

Hong Kong Hospital Authority Hong Kong S.A.R., China
Data Scientist in AI & Data Collaboration Lab Oct 2023 – Aug 2024

- Built **LangChain LLM tool** that locates relevant patient-record information and drafts medical reports for physician review, reducing report-analysis latency by **70%**; later deployed at **6 public hospitals**, where AI processes ∼80% of medical reports and saves ∼40% of staff paperwork time (The Standard; HKFP, 2025)

Hong Kong Applied Science and Technology Research Institute Hong Kong S.A.R., China
Research Engineer in Smart Mobility Technologies Group Jan 2023 – Oct 2023

- Reproduced and extended academic research for product development, resulting in **patent application** (Google Patent)
- Developed Flask backend integrations between smart-mobility simulators (**CARLA and SUMO**) via REST APIs; contributed bug fixes and features to open-source OpenCDA and OpenCOOD projects

Siemens Ltd. Hong Kong S.A.R., China
Software Engineering Intern in Department of Smart Infrastructure Jun 2020 – Jun 2021

- Built internal energy-consumption CMS with **MEAN stack**, hosted via AWS (EC2, S3); onboarded **100+ users**
- Trained **time series forecasting** models (RNN, LSTM) for **anomaly detection** on HVAC machine performance, achieving **74% F1** and outperforming rule-based engineering baseline by 17%; visualized results with Matplotlib

Honors & Awards

First Class Honors (awarded for CGPA $\geq$ 3.50/4.30), **City University of Hong Kong** 2022

Grand Challenges Scholar, **U.S. National Academy of Engineering** (with CityU program scholarship) 2022

College of Engineering Mentor Service Award, **City University of Hong Kong** 2022

Innovation and Technology Scholarship (CityU News), **HKSAR Government and HSBC** 2021

Fung Scholarship, **Victor and William Fung Foundation** 2021

Shanghai Commercial Bank Scholarship, **Shanghai Commercial Bank** 2021

Microware Scholarship, **Microware Limited** 2021

Teaching & Service

Reviewer, **CoRL** 2026; **AAAI** 2026, 2027 2025 – 2026

Mentor, Sustainability Development Mentorship Programme, **Hong Kong Federation of Youth Groups** 2022

Mentor, Student Mentoring Scheme for first-year students, **City University of Hong Kong** 2019 – 2022

Committee Member, Google Developer Student Club, **City University of Hong Kong** 2020 – 2021

Teaching Assistant, CS2312 Problem Solving and Programming in Java, **City University of Hong Kong** Fall 2020